Peter Chovanec, PhD

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Education

2014 – 2019	**PhD**	University of Cambridge	Babraham Institute
2010 – 2014	**BSc**	University of York	Genetics with a research Year in Europe

Research experience

2025 Nov-present — Assistant Project Scientist, University of California, Los Angeles
Department of Human genetics – Prof. Yi Yin

- *Project:* Application of sci-L3-Strand-seq on over 200 samples across multiple projects investigating DNA repair mechanisms, including single-strand break repair involving PARP genes and PARP inhibitors; establishing *error-free* DNA repair as a tumor "mutation" signature to guide treatment selection; and defining the role of error-free repair in in rare hereditary cancer and cancer-predisposition syndromes.

2020 Nov-2025 Oct — **Postdoctoral scholar**, University of California, Los Angeles
Department of Human genetics – Prof. Yi Yin

- *Project 1:* Extension of the sci-L3 combinatorial indexing methods to template strand sequencing (sci-L3-Strand-seq), genome conformation capture (sci-L3-Hi-C), and the joint profiling of RNA and chromatin accessibility (sci-L3-RNA/ATAC).
- *Project 2:* Development of novel computational approaches for the comprehensive capture of sister chromatid exchanges and structural variations from low-depth single-cells template strand sequencing (sci-L3-Strand-seq).

2019 Mar-2020 Nov — **Postdoctoral scholar**, California Institute of Technology
Division of Biology and Biological Engineering - Prof. Mitch Guttman

- *Project 1*: Development of methods and computational tools to examine the higher-order structure of RNA and chromatin within the nucleus (RD-SPRITE)
- *Project 2*: Development of novel single-cell RNA-seq methods utilising the SPRITE combinatorial indexing scheme for the simultaneous readout of RNA and DNA

2014 Oct-2019 Feb — **PhD**, University of Cambridge, Babraham Institute
Department of Nuclear dynamic - Dr. Anne Corcoran

- *Project 1*: Investigating B cell development and antibody repertoires in a humanised mouse model and method development for unbiased quantification of antibody repertoires (VDJ-seq)
- *Project 2*: Examining the heterogeneous transition during primed to naïve reprogramming of human pluripotent stem cells (collaboration with Dr. Peter Rugg-Gunn)
- *Project 3*: Investigating the remodelling of the 3D genome organization between naïve and primed human pluripotent states (collaboration with Dr. Peter Rugg-Gunn and Dr. Stefan Schoenfelder)
- *Project 4*: Assembly of the immunoglobulin heavy chain locus of the 129sv mouse strain utilising the Nanopore Minion long-read sequencing

2013 Oct-9 months — **BSc dissertation**, University of York
Department of Biology – Prof. Kanchon Dasmahapatra

- *Project*: Examining the feasibility of using the Genome Analysis Toolkit (GATK) for accurate SNP genotyping in non-model organisms (*Heliconius* butterfly).

2012 Sep-10 months — **Research placement**, IGBMC, Strasbourg
Department of Functional Genomics and Cancer - Dr. Hinrich Gronemeyer

- *Project*: Elucidating the differential effects of the CREB-binding protein (CBP) versus its paralogue p300 in the context of estrogen receptor alpha (ERα) mediated transcription.

Publications

1. **Chovanec, P.**, He, S. & Yin, Y. (2026) Genome-wide mapping of Cas9-induced sister chromatid exchange across single and over 200 genomic targets. Elife 10.7554/elife.112073.
2. He S., **Chovanec, P.**, Yin, Y. (2026) Protocol for mapping sister chromatid exchanges and structural variations using sci-L3-strand-seq. STAR protocols 7 (4).
3. **Chovanec, P.**, Ridgley, T. & Yin, Y. (2026) High-throughput mapping of spontaneous mitotic crossover and genome instability events with sci-L3-Strand-seq. Nucleic Acids Res. 54, gkag119.
4. White, H.N., **Chovanec, P.**, Biggins, L., French, E.C., Bullen, G., Andrews, S., and Corcoran, A.E. (2025). VH-replacement shapes the antibody repertoire by removing the genes of non-functional heavy-chains. EMBO J., 1–21.
5. **Chovanec, P.**, Yin, Y. (2025) Generalization of the sci-L3 method to achieve high-throughput linear amplification for replication template strand sequencing, genome conformation capture, and the joint profiling of RNA and chromatin accessibility. Nucleic Acids Res. 53, gkaf101.
6. Quinodoz, S.A., Bhat, P., **Chovanec, P.**, Jachowicz, J., Ollikainen, N., Detmar, E., Soehalim, E., Guttman, M. (2022) SPRITE: A genome-wide method to map higher-order 3D spatial interactions in the nucleus using combinatorial split-and-pool barcoding. Nat Protoc 17, 36–75.
7. **Chovanec, P.** & Yin, Y. A mapping platform for mitotic crossover by single-cell multi-omics. (2021) Methods Enzymol 661, 183–204.
8. Quinodoz, S.A., Jachowicz, J.W., Bhat, P., Ollikainen, N., Banerjee, A.K., Goronzy, I.N., Blanco, M.R., **Chovanec, P.**, Chow, A., Markaki Y., Thai, J., Plath, K., Guttman, M. (2021) RNA promotes the formation of spatial compartments in the nucleus. Cell *184*, 1–16.
9. **Chovanec, P.†**, Collier, A.J.†, Krueger, C., Várnai, C., Semprich, CI., Schoenfelder, S., Corcoran, A.E., Rugg-Gunn, P. (2021) Widespread reorganisation of pluripotent factor binding and gene regulatory interactions between human pluripotent states. Nat Commun *12*, 2098.
10. Teng, Y.*, Young, J.L.*, Edwards, B.*, Hayes, P., Thompson, L., Johnston, C., Edwards, C. Sanders, Y., Writer, M., Pinto, D., Zhang,Y., Roode, M., **Chovanec, P.**, Matheson, L., Corcoran, A.E., Fernandez, A., Montoliu, L., Rossi, B., Tosato, V., Gjuracic, K., Nikitin, D., Bruschi, C., McGuinness, B., Sandal,T., and Romanos, M. (2019) Diverse Human VH antibody fragments with bio-therapeutic properties from the Crescendo Mouse. New Biotechnol *55*, 65–76.
11. Koohy, H.†, Bolland, D.J.†, Matheson, L.S.†, Schoenfelder, S., Stellato, C., Dimond, A., Varnai, C., **Chovanec, P.,** Chessa, T., Denizot, J., Garcia, R.M., Wingett, S.W., Freire-Pritchett, P., Nagano, T., Mielczarek, O., Baizan-Edge, A., Stubbs, B.A., Hawkins, P., Stephens, L., Elderkin, S., Spivakov, M., Fraser, P., Corcoran, A.E., and Varga-Weisz, P.D. (2018) Genome organisation and chromatin analysis identifies transcriptional downregulation of insulin-like growth factor signalling as a hallmark of ageing in developing B cells. Genome Biol. *19*, 126.
12. **Chovanec, P.**, Bolland, D.J., Matheson, L.S., Wood, A.L., Corcoran, A.E. (2018). Unbiased quantification of immunoglobulin diversity at the DNA level with VDJ-seq. Nat. Protoc. *13*, 1232–1252.
13. Collier, A.J.†, Panula, S.P.†, **Chovanec, P.***, Schell, J.P.*, Reyes, A.P., Petropoulos, S., Corcoran, A.E., Walker, R., Douagi, I., Lanner, F., Rugg-Gunn, P.J. (2017). Comprehensive Cell Surface Protein Profiling Identifies Specific Markers of Human Naive and Primed Pluripotent States. Cell Stem Cell *20*, 874–890.e7
14. Matheson, L.S., Bolland, D.J., **Chovanec, P.**, Krueger, F., Andrews, S., Koohy, H., and Corcoran, A. (2017). Local chromatin features including PU.1 and IKAROS binding and H3K4 methylation shape the repertoire of immunoglobulin kappa genes chosen for V(D)J recombination. Front. Immunol. *8*, 1550.
15. Bolland, D.J., Koohy, H., Wood, A.L., Matheson, L.S., Krueger, F., Stubbington, M.J.T., Baizan-Edge, A., **Chovanec, P.**, Stubbs, B.A., Tabbada, K., Andrews, S.R., Spivakov, M., Corcoran, A.E. (2016). Two Mutually Exclusive Local Chromatin States Drive Efficient V(D)J Recombination. Cell Rep. *15*, 2475–2487.

Equal contributions: *; Co-first author: †

Manuscripts under review

1. **Chovanec, P.**, He, S. & Yin, Y. (2026) Lesions initiating spontaneous mitotic crossover are minimally subject to non-homologous end joining. bioRxiv 2026.05.11.724412 (minor revision in *Nucleic Acids Res.*)
2. Ridgley T., **Chovanec, P.**, He S., Yin Y. Robust Chromosome Segmentation in Sparse Single-Cell Whole-Genome Sequencing Using a k-Segment Constrained Hidden Markov Model. bioRxiv 2026.10.19.760240 (Submitted to *Nucleic Acids Res.*)

3. Collier, A.J., **Chovanec, P.**, Yue, S., Krall, A.S., Heinkele, F., Sha, J., Feng, B., Ashley, I., Sims, R., Rubbi, M., Langerman, J., Hwang, Y., Clark, A.T., Pellegrini, M., Bouhaddou, M., Christofk, H., Yin, Y., Wohlschlegel, J., Jonas, S., Plath, K. (2026) Pluripotent state-specific mechanisms orchestrate human blastoid development.

Manuscripts in preparation

1. Collier, A.J.*, **Chovanec, P.***, Shrestha, P.*, Clémot, M.*, Sun, Y., Susser, S., Feng, B., Krall, A.S., Bendall, A., de Sousa, J.A., Fabian, C., Christofk, H., von Meyenn, F., Rugg-Gunn PJ, Yin, Y., Plath, K. X chromosome dosage controls human embryonic and extraembryonic development
2. **Chovanec, P.***, He, S.*, Yin Y. Scalable and comprehensive mapping of crossover outcomes in rare disease patient-derived DNA repair gene variants reveals non-canonical sister chromatid exchange pathways. (work presented at the 2026 FASEB meeting, where it received a Best Poster Award, and provided the foundational preliminary data for the lab's recently funded W. M. Keck Foundation Award, $250K annual DC 2026-2028).
3. **Chovanec, P.***, He, S.*, Yin Y. Genetic regulation of error-free sister chromatid exchange and mutational structural variations induced by chronic endogenous versus acute pharmacologic PARP trapping. (work presented at the 2026 GRC meeting, where it received a 2nd place Poster Award, and provided the foundational preliminary data for the lab's recently funded DOD CDMRP Idea Development Award, $2.1M DC+IDC 2026-2030).

Honours, awards, invited talks

2026	Poster Award (4 selected), FASEB Genetic Recombination and Genome Rearrangements, AZ
2026	2nd Place Poster Award (93 international entries), GRC DNA Damage, Mutation and Cancer, CA
2023, 2024	Invited talk, SoCal Genome Stability Symposium, CA
2022	Poster Presentation Award, SoCal Genome Stability Symposium, CA
2022	Invited talk, FASEB Genetic Recombination and Genome Rearrangements, CO
2021	Invited talk, SoCal Genome Stability Symposium, CA
2017	Babraham Institute Knowledge Exchange Commercialisation travel award, UK
2017	University of Cambridge Sidney Sussex College travel award, UK
2014-2019	Industrial CASE studentship, Babraham Institute, UK
2010-2014	The York Award, University of York, UK

Select Poster Presentations

2024	GRC DNA Damage, Mutation and Cancer, Ventura, USA
2021-2024	SoCal Genome Stability Symposium, Los Angeles
2018	Poster at Keystones Symposium: Chromatin Architecture and Chromosome Organization, Canada
2016	Poster at Single Cell Genomics meeting, Sanger Institute, UK

Teaching and professional experience

2018-present	Peer-reviewer for Frontiers in Immunology, Cell Reports Methods, and Epigenetics & Chromatin.
2018-present	Postdoc, Graduate and Undergraduate students supervised: Sam Rees (Babraham Institute, graduate, VDJ-seq methodology and analysis), Yujing Yang and Paulomi Bhattacharya (Caltech, graduate, single-cell technology development), Trevor Ridgley (UCLA, graduate, computational method development) Jessica Javaherforoush, Chizaram Okeiyi, and Marat Zargaryan (UCLA, undergraduate, CRISPR-Cas9 genotyping) Shiyang He (UCLA, postdoc, sci-L3-Strand-seq applications)
2017	Public engagement exhibit designer (Making your Mark: Exploring Epigenetics)
2016	Symposium of Biological and Life Sciences (SymBLS) conference coordinator
2015, 2016	Public engagement exhibit designer and presenter at the Royal Society Exhibition and the Cambridge Science festival (Immune Army: Weapons of Microscopic Destruction)